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PI Name:	Gerhart, John C Ph.D.		
Project Title:	Hemichordates and the Origin of Chordates		
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Space Biology Special Category:	None		
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	The aim of the research is to compare the development and organization of hemichordates to the development and organization of chordates, thereby to discern the origin of chordate traits in non-chordate ancestors. Hemichordates are the phylum of bilateral animals closest to chordates, the phylum to which vertebrates belong, including humans. Hemichordates have long been thought to possess chordate likenesses and therefore to hold value for identifying chordate traits that were present in ancestors predating chordates. However, the anatomy of these three-part burrowing marine worm-like animals is in many ways unlike chordates, and controversy has surrounded the anatomical comparisons of the past century. Our studies (NASA funding) have entailed the use of modern molecular methods of comparison, particularly in situ hybridization. Thus far, with probes to chordate neural patterning genes, we have assessed hemichordate-chordate similarities of the nervous system. The domains of expression of neural patterning genes have been extensively mapped in the chordate nervous system. At this time, 28 orthologous genes have been used		

Task Description:	from hemichordates, and the conclusion is strongly supported that the anterior-posterior dimension of the hemichordate nervous system is very similar to that of chordates in terms of expression domains of neural patterning genes. This close similarity to chordates was not previously known. The topography of domains is strikingly alike even though the hemichordate nervous system is not centralized as in chordates but is diffuse--a basiepithelial nerve net with tracts of axons along the dorsal and ventral midlines. As a speculation on origins, the domain topography may have arisen in an ancestor with a diffuse nerve net, and the topography was then carried into the chordate nervous system as it was centralized. In the dorsoventral dimension, the hemichordate is more like a protostome (e.g., Drosophila) than a chordate. It has a Bmp-Chordin axis in the Drosophila orientation. Slit and Netrin at the dorsal and ventral midlines respectively pattern axon migration to form the two axon tracts. Our first paper was published last year; another is in preparation. A White Paper prepared by us was accepted, leading to the preparation of a BAC library for S. kowalevskii. A White Paper prepared by us with members of Eric Lander's laboratory has received high priority from NHGRI for sequencing of the entire genome within the next year. The objective of the research is to determine which chordate traits were already present in a non-chordate ancestor shared with hemichordates. The traits include modes of development, domains of expression of genes, molecular interactions, physiology, and morphology. Knowledge about the ancestry of traits, especially molecular and developmental traits dating back to pre-Cambrian ancestors with other phyla, gives insights about what is fundamental and conserved in chordates and what was available to the chordate line for its unique evolutionary innovations. From our studies thus far, the hemichordate seems to have many basic chordate-like characters, but in a less complicated context where processes and components may be easier to elucidate.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Knowledge about the origin of our phylum, the chordates, including ancestry of the body plan, cell types, developmental modes, and molecular components and interactions.

This is the same as the Annual Report
Grant number: NAG2-1605 . Principal Investigator: John Gerhart Institution’s name: University of California, Berkeley
Period of report: 10/9/03-10/8/04

Brief description of goals/objectives: The aim of the research is to compare the development and organization of hemichordates to the development and organization of chordates, thereby to discern the origin of chordate traits in non-chordate ancestors. Hemichordates are the phylum of bilateral animals closest to chordates, the phylum to which vertebrates belong, including humans. Hemichordates have long been thought to possess chordate likenesses and therefore to hold value for identifying chordate traits that were present in ancestors predating chordates. However, the anatomy of these three-part burrowing marine worm-like animals is in many ways unlike chordates, and controversy has surrounded the anatomical comparisons of the past century. Our studies (NASA funding) have entailed the use of modern molecular methods of comparison, particularly in situ hybridization and siRNA knock down of gene expression. Thus far, with probes to chordate neural patterning genes, we have assessed hemichordate-chordate similarities of the nervous system. The domains of expression of neural patterning genes have been extensively mapped in the chordate nervous system. At this time, 28 orthologous genes have been used from hemichordates, and the conclusion is strongly supported that the anterior-posterior dimension of the hemichordate nervous system is very similar to that of chordates in terms of expression domains of neural patterning genes. This close similarity to chordates was not previously known. The topography of domains is strikingly alike even though the hemichordate nervous system is not centralized as in chordates but is diffuse--a basiepithelial nerve net with tracts of axons along the dorsal and ventral midlines. In the past year we have obtained many striking in situ patterns and have developed methods to visualize internal expression of these genes. As a speculation on origins, the domain topography may have arisen in an ancestor with a diffuse nerve net, and the topography was then carried into the chordate nervous system as it was centralized. Our first article was been published in 2003 and a further one is in preparation. A White Paper submitted by us for BAC library preparation was accepted two years ago and the library is now available. A White Paper prepared by us with members of Eric Lander's laboratory has received high priority from NHGRI for sequencing of the entire genome within the next year.

Brief summary of last year’s progress: Last year I described our progress on the anteroposterior map of domains of expression of neural patterning genes in hemichordates, compared to chordates. The maps are very similar despite the fact that chordates have a central nervous system whereas hemichordates have a non-centralized basiepithelial nervous system. This year I will report on our progress in comparing the dorsoventral organization of hemichordates and chordates, with further insights into the ancestry of the nervous system.

The Bmp-Chordin polarity: In both chordates and Drosophila, Bmps (or the Dpp ortholog) and Bmp-antagonists (Chordin or Sog) are situated at opposite poles of the body plan for dorsoventral patterning. To our delight, in hemichordates this arrangement holds as well: the dorsal midline of the hemichordate’s embryonic ectoderm is a hotspot of Bmp signaling. Running the embryo’s length is a thin stripe of expression of bmp2/4, bmp7, twg, crossveinless, and bambi/nma. Twg and Crossveinless are thought to be Bmp co-ligands involved in Bmp’s intercellular transport. Bambi/nma is a curious Bmp receptor fragment (a natural dominant negative) that inhibits Bmp receptor function in the cell in which it is expressed. The ventral midline of the ectoderm of the hemichordate is the site of chordin gene expression, in a rather broad stripe (90° of circumference). The polarity may be more like that of Drosophila (protostomes) since both bmp and chordin are expressed in the ectoderm and in a protostome polarity (Bmp dorsal), whereas in chordates the chordin is expressed initially in the notochord (dorsal endomesoderm, not ectoderm) and Bmp is ventral in the ectoderm.

Dorsoventral domains: Several genes (netrin, sim, nkx2.2[vnd], gsx[ind], and msx[msh]) are expressed in both the Drosophila and chordate nervous systems, and in the same domain order, even though the nerve cords are rather different. The chordate system is a dorsal, hollow, basiepithelial cord, with sensory neural crest derived from nearby, whereas the Drosophila version is a ventral, solid, ladder-like, sub-epithelial cord with sensory neurons derived from a distant dorsal ectoderm. In chordates, at least 40 more genes are expressed in dorso-ventral stripes in the hindbrain and spinal cord, such as nkx6.1, olig2, pax6, en, dbx, and islet. In the ventral midline ectoderm of the hemichordate embryo, netrin, and sim are indeed expressed, as in the Drosophila ventral midline and as in the floorplate of the chordate neural tube. The role of Netrin in the hemichordate ventral midline seems clear in light of the work of others on Drosophila, nematodes, and chordates. Namely, a Netrin gradient probably spreads laterally from the midline, attracting some axons it to form the ventral axon tract, while repelling others to form the dorsal axon tract. Consistent with this is old evidence that the ventral tract is primarily a motor tract. Most of the striated muscle of the hemichordate is in the ventral half of the body, close to the putative motor tract. In the dorsal midline of the hemichordate, slit is expressed, also known to

Task Progress:	attract or repel axons of various kinds. Thus, Netrin ands Slit gradients seem to coordinate the dorsoventral axon pattern of the diffuse nervous system. The axon tracts presumably connect distant sensory inputs and motor/visceral/secretory outputs. Despite the netrin, sim similarity, we do not find nkx2.2, gsx, or msx expressed in dorsoventral domains in hemichordates (making them different from both Drosophila and chordates), nor do we find nkx6.1, olig2, pax6, en, dbx, or islet in dorso-ventral domains like those of the chordate neural tube (though they have full-fledged chordate-like antero-posterior expressions of these genes). In the chordate neural tube, many of these genes are activated and repressed by a double gradient of Shh in the floorplate and Bmp in the roofplate. To our surprise, hemichordates do not express the hedgehog gene anywhere on the ventral midline, but only at the anterior tip of the embryo. We think the hedgehog ligand and these hedgehog-dependent domain may have been co-opted in the chordate line, independently from protostomes, in the course of the centralization of interneurons in the nervous system. Hemichordates do show several chordate-like dorsoventral domains, ones not shared by Drosophila. Admp, for example, is expressed strongly at the ventral midline. Its expression is well known in chordates in the notochord (along with Chordin) under the floorplate of the neural tube. Admp is the “anti-dorsalizing morphogenetic protein”, an interesting but Bmp3-relative, that in chordates not only antagonizes Bmp2, 4, and 7, but also represses the head organizer. Likewise, dlx and tbx2/3 orthologs are expressed in hemichordates on the dorsal midline in a wide ectodermal stripe (60° of circumference), spanning the Bmp stripe. In chordates these Bmp-dependent genes are expressed in neural crest at the edges of the neural plate, close to the Bmp-secreting roofplate. Thus, several chordate-like signals and responses are superimposed on the protostome-like dorso-ventral plan. Success with siRNAs: We have recently been able to knock out bmp expression in the hemichordate embryo using dicer siRNAs and specific single siRNAs. Over 100 eggs have been injected and many in situ studies will be done. The phenotype is striking—the embryo develops as a cylindrical ventral embryo, with no trace of dorsal morphological traits. That is, the metasome does not bend ventrally but remains straight; the telotroch remains symmetrical and circumferential, the gill slit pair does not form in a dorsolateral position, and finally , as the ventral mouth forms, the proboscis falls off, as if the mouth opens all the way across. This opens a new ear of functional studies. In preliminary studies, siRNAs for brachyury, six3, and otx have also been found to give consistent phenotypes with defects at the particular sites of expression. Conclusions: The dorso-ventral dimension of the hemichordate looks rather like a protostome such as Drosophila in its expression of genes for Bmp (Dpp) signals in the dorsal midline and for Bmp antagonists (Chordin or Sog), as well as for Netrin and Sim, in the ventral midline. In chordates, the initial Bmp-Chordin signals are thought to be involved in a neural-epidermal choice (the neural default model), but Bmp also has a later neural role (opposite Shh) in patterning sensory neuronal cell types in the neural tube. In Drosophila the Dpp-Sog opposition in the body wall ectoderm patterns sensory versus motor/interneuron fates, rather than a neural vs. non-neural alternative. We hypothesize that hemichordates are rather like the Drosophila case: neurogenesis occurs everywhere in the ectoderm, but perhaps with more sensory neurons arising in the dorsal ectoderm and more motor neurons/interneurons in the ventral ectoderm. Interneurons, we think, may be expressed both dorsally and ventrally, less centralized than in Drosophila or chordates. Presumably the deuterostome ancestor had a Bmp-Chordin axis and a diffuse nervous system, which was later centralized in the chordate line by introducing a Default Mechanism for repressing all neurogenesis in epidermal ectoderm by Bmp. The neural tube then continued the Bmp-based sensory-motor patterning, while adding the entire Shh-based patterning (which is entirely absent from Drosophila and hemichordates) with a small assist from chordin (to diminish Bmp in the ventral tube). Awards, publications, presentations: Publications: Gerhart, J. (2004) Symmetry breaking in the egg of Xenopus laevis. Pp. 341-351. In “Gastrulation” (Edited by CD Stern) Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. Lowe, C.J., Tagawa, K., Humphreys, T. Kirschner, M., and Gerhart, J. (2004) Hemichordate embryos; procurement, culture, and basic methods. pp. 207-219. In: Methods in Cell Biology. G.A. Wray, C. Ettensohn, and G. Wessel (eds). San Diego: Elsevier Science. Presentations: J. Gerhart presented this work at the: Sloan-Kettering Symposium on Developmental Biology, New York, October 17-18, 2003, and Cold Spring Harbor Symposium on Evolution and Development, Cold Spring Harbor, N.Y., March 31-April 3, 2004. New Technology: None Future research plan: We will more fully characterize the diffuse but well patterned nervous system of hemichordates. Mnx (the motor neuron homeobox) is expressed ventrally, we know, and poxn seems dorsally expressed. We have started with neuron-specific antibodies and confocal microscopy, finding that serotonin-producing neurons arise in the dorsal ectoderm and send long axons mostly to the ventral nerve tract rather than the dorsal tract. We will use antibodies against other neurotransmitters, and we have many excellent neuron specific gene sequences to test for localized expression, such as tau, brn3, brn1/2/4, contactin, NrCAM, nato, math6, elav, musashi/nrp, and sox2/3. Also, we will begin characterizing the main signaling centers in the ectoderm of the embryo, such as the anterior tip signaling center (resembling the anterior neural ridge of chordates), the prosome base center, the first gill slit center (resembling the midbrain-hindbrain organizer of chordates), and the posterior tip. These are sites of secretion of various Wnt and FGF ligands, which presumably pattern nearby embryonic regions. Furthermore, we will characterize the dorsal and ventral midlines of the embryonic gut in which are expressed a variety of orthologs of genes expressed in Spemann’s organizer in chordates. In pursuing these plans, we will use siRNAs and various inhibitors and antagonists to test the developmental function of the signaling centers, in comparison to chordates. Significant budget of key personnel changes: None
Bibliography Type:	Description: (Last Updated: 04/02/2007)
Books/Book Chapters	Gerhart, J. "Symmetry breaking in the egg of Xenopus laevis." in "Gastrulation: From Cells to Embryo." Ed. C.D. Stern. Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y., p. 341-351, 2004. , Mar-2004

Books/Book Chapters	Lowe, C.J.; Tagawa, K.; Humphreys, T.; Kirschner, M.; Gerhart, J. "Hemichordate embryos; procurement, culture, and basic methods." in "Development of Sea Urchins, Ascidians, and Other Invertebrate Deuterostomes: Experimental Approaches (Methods in Cell Biology, Vol. 74)." Ed. G. Wray, C.E. Ettensohn, and G.M. Wessel. Academic Press/Elsevier. San Diego CA., p. 207-219, Oct 2004. , Oct-2004